

Drug3D-DTI: Improved Drug-target Interaction Prediction by Incorporating Spatial Information of Small Molecules

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Abstract—A number of machine learning (ML) approaches for drug discovery have been available that rely only on sequential (1D) and planar (2D) information without effectively using the 3D information for generating features of drugs. However, 3D information of small molecules can reflect relative position of atoms more directly, which affects molecular properties. In this work, we present a new deep learning model called Drug3D-DTI for drug-target interaction prediction. Drug3D-DTI takes advantage of molecular spatial information, i.e., atom proximity in three-dimensional (3D) structures. We comprehensively evaluated the performance of Drug3D-DTI on two datasets with two tasks of regression and classification. In particular, we compared Drug3D-DTI with several existing methods including the two cutting-edge methods for compound-protein interaction prediction. From the experimental results, Drug3D-DTI clearly outperformed other methods under all settings. Further, this performance improvement was validated by ablation experiments and a case study. The implementation of Drug3D-DTI is available at (<https://github.com/zhiruiliao/Drug3D-DTI>).

Index Terms—drug-target interaction, 3D structure of small molecule, graph neural network, Transformer

I. INTRODUCTION

In the field of drug discovery, a critical issue is how to characterize drug-target interactions for screening potential drug candidates to accelerate drug development. For improving the efficiency and reducing the cost, two computational approaches, docking simulation [1], [2] and machine learning (ML) [3]–[5], have been widely applied to address this issue. Compared with docking simulation, machine learning-based

approaches usually use fewer computing resources, without requiring the availability of the 3D structural information of target proteins. As such, ML-based approaches for drug discovery becomes increasingly important in the last few years.

Specifically, a variety of ML methods for drug-target interaction (DTI) prediction have been proposed by extracting information from input data with different sources. According to what kind of data they use, these methods can be classified into two types: 1) the use of the internal information of drugs and targets, such as the drug molecular graph and the amino acid sequence of a target [6]–[13]; 2) the use of the external information, such as drug-drug interaction, drug side effect and protein-protein interaction networks, as well as drug-disease and target-disease associations [14]–[19]. The first-type methods further have two main categories: similarity based and feature (or representation) based methods. Similarity-based ML methods [4]–[7] have one key underlying assumption that similar targets tend to share similar drugs and vice versa. The most widely used similarity metric among drugs is Tanimoto coefficient, which computes the structural similarity between the fingerprints of two molecules. Meanwhile, sequence-based similarity, such as Smith–Waterman score [20], has been widely used in the target side. A notable weakness of similarity-based ML methods is that computing similarities among all molecules becomes intractable when the number of small molecules is huge. On the other hand, feature-based ML methods [8]–[13], especially deep learning based methods in an end-to-end style, can deal with data of large scale, where encoders extract high-level features of drugs and targets, and then decoders make predictions.

In a word, the first-type methods could be widely applied as long as the basic information of drugs and targets is available. In contrast, the second-type methods rely heavily on

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the availability of extensive knowledge about drugs and targets [15]–[19], which is generally unknown in practice, particularly for a novel drug or target proteins. In this work, therefore we focus on feature (representation) based methods of the first-type methods by using the internal information of drugs and targets only.

Extracting effective drug features (or learning effective representation) is a crucial step in ML-based DTI methods [21]. Currently, these features are extracted mainly from two sources: 1) one-dimensional (1D) information, such as molecular fingerprints and SMILES (strings) [8], [9], [13], and 2) two-dimensional (2D) information such as molecular graphs [10]–[12]. Generally, molecular fingerprints are derived from manual feature engineering, which may lose information; and a molecule always corresponds to more than one dissimilar SMILES strings, which introduces instability. Such properties of 1D information limit their application. However, drug-target interaction is caused by the binding between small drug molecules (ligand) and target proteins like hand-in-glove. This binding process is determined by both functional groups (which depend on 2D chemical bonds) and the 3D structure of drugs. In particular, 3D structures of drugs (small molecules) where each atom in a drug has its position in 3D space affect their properties through steric effects and hydrogen bond, for example. In fact, atoms that are far away from each other in 2D structures may be close in their 3D structures. As an example, Fig. 1 shows that nitrophenol isomers share the similar chemical bond arrangements. In 2-nitrophenol, an oxygen atom in the nitro group is away from the hydrogen atom in the hydroxyl group on the molecular graph. But they are close in their 3D structure. This fact results in a stronger intramolecular force (H-bond) in 2-nitrophenol than in 3-nitrophenol and 4-nitrophenol, as well as a much lower melting point of 2-nitrophenol than the other two isomers.

On one hand, both 2D and 3D information of drugs are vital for drug-target interaction prediction. 3D features cannot be directly derived from 1D and 2D information. On the other hand, without 3D structure information of targets, we still can model the 3D structures of drugs by using cheminformatics software without consuming much computation. This is because drug compounds contain only a small number of atoms. As such, it is possible to incorporate both 2D and 3D information of drugs for learning their effective representations.

In this work, we propose a model called Drug3D-DTI that can learn more effective drug features for drug-target interaction prediction by using both their 2D and 3D information. Specifically, using the bond and features of the 2D molecular graph of a drug, as well as the 3D spatial features of its 3D structure, Drug3D-DTI relies on two graph convolutional networks (GCNs) and fully connected layer(s) to produce the final feature matrix of the drug for downstream tasks of regression and binary classification. The performance of Drug3D-DTI has been empirically validated under several settings including classification and regression on different datasets. The experimental results have demonstrated that Drug3D-DTI outperformed the compared methods under all

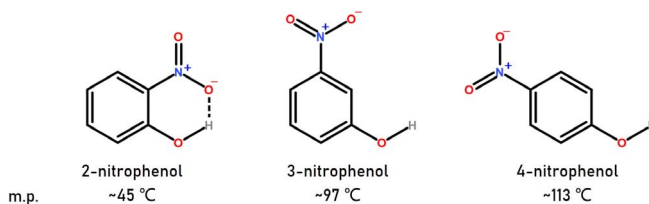


Fig. 1. Nitrophenol isomers. The numbers under the isomer names are melting points (m.p.) [22]. The dashed line of 2-nitrophenol is an H-bond, which results in an intramolecular force of 2-nitrophenol stronger than that of the other two isomers.

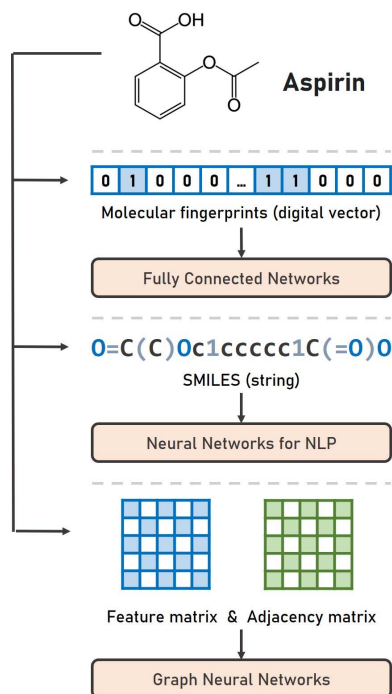


Fig. 2. Three types of drug (small molecule) feature learning: Fingerprints and SMILES are one-dimensional (1D) representations while adjacency matrices reflect two-dimensional (2D) features.

settings. This conclusion was further confirmed by ablation experiments and a case study.

In summary, the contributions of this work are as follows:

- For drug-target interaction prediction, we propose an ML model called Drug3D-DTI that uses both 2D and 3D information of drugs and the amino acid sequence of targets.
- By experiments, we demonstrate that incorporating 3D information of drugs can improve drug-target interaction prediction on both regression and binary classification by comparing with several state-of-the-art baselines.

II. RELATED WORK

Based on drug molecular representations, we briefly review three types of the existing methods for drug-target interaction prediction with a focus on GCN. 1) **molecular fingerprints**, such as Extended-Connectivity Fingerprints (ECFP) [23]: a pre-defined vector [24] as a direct input of machine learning

models [9], [25], [26]. 2) **string representation** like SMILES [27]: ASCII strings processed by pipelines that are similar to natural language processing [8], [13], [28], [29]. Both fingerprints and strings are considered as 1D information, while a molecule can be regarded as a graph with nodes for atoms and edges for chemical bonds, each of which connects two atoms. 3) **molecular graph** or 2D topology: an undirected graph where nodes and edges indicate atoms and bonds, respectively. With the use of graph neural networks, especially graph convolutional network (GCN) [30], many studies applied GCN to bioinformatics [31]–[33], such as drug representation in drug-target interaction prediction [10]–[12] by taking molecular graphs as the input. Accepting a graph (the adjacency matrix of nodes) with features assigned to nodes, GCN aggregates features from its neighboring connecting nodes for each node, achieving excellent performance on learning representation encoded in a given graph with initial features. However, graphs that are directly derived from chemical bond connectivity sometimes may not reflect real neighbor relationships between the atoms. For example, in 2-nitrophenol of Fig. 1, an oxygen in the nitro group and the hydrogen in the hydroxyl group are spatially close, generating an H-bond. In contrast, the distance between these two nodes in their molecular graph is far from the five edges. This means that at least five layers are required for GCN to capture the interaction between these two atoms. Thus we believe that real distances between atoms, i.e. spatial or 3D distances, would be the additional key information to establish a graph for extracting better features via GCN.

As a summary, Fig. 2 illustrates the three types of drug molecular representations for drug-target interaction machine (deep) learning methods.

Except for the drug molecular representations, the system architecture is also important for predicting drug-target interaction. Classical architecture usually has the dual-tower model, which comprises of two encoders for representation learning of respective drug and target features and then concatenates them to make predictions via fully connected layer(s). Also, the Transformer [34] architecture, which is originally designed for machine translation via attention mechanisms, has been applied to solve bioinformatics problems, with great success. In particular, DeepDTA [8] and TransformerCPI [11] are typical models of state-of-the-art deep learning architectures. Specifically, the input of DeepDTA is a string (SMILES) for a compound and an amino acid sequence for a protein, which are encoded into feature vectors of the molecule and protein by 1D-convolutional layers and a max-pooling layer, respectively. These two vectors are concatenated by the output module as the final vector, which serves as the input of fully connected layers for prediction. Unlike DeepDTA, TransformerCPI represents a compound by both a bond matrix and an atom feature matrix, as well as a protein by a set of 3-gram amino acid “words”. They are then encoded by graph convolutional network and convolution with the gated linear unit (GLU) [35], respectively. The output module begins with the Transformer decoder to produce a feature vector, which is finally inputted

into fully connected layers for prediction.

In our experiments, we replace the encoded vector for drugs used in DeepDTA and TransformerCPI with that from Drug3D-DTI. As such, this replacement can be used for evaluating the representation capacity of Drug3D-DTI on the same architecture of the system.

III. METHOD

In this section, we present our proposed Drug3D-DTI for two drug-target interaction prediction.

A. Drug Encoder

1) *Generating Drug Input Matrices*: Let n be the number of nodes (atoms) and f be the size of the initial (binary) feature vector of each atom. Drug3D-DTI has three input matrices: 1) **feature matrix** $X \in \mathbb{R}^{n \times f}$: By following [11], we obtain binary atom (node) feature vectors ($f = 34$) according to chemical features listed in Table I via RDKit, an open-source cheminformatics software library (<https://www.rdkit.org>). 2) **bond matrix** $A \in \mathbb{R}^{n \times n}$: This is equivalent to a molecular graph (used in the existing work of [10]–[12]), directly generated from covalent bonds. That is, A_{ij} equals 1 if there is a covalent bond between atom i and atom j , and 0 otherwise. 3) **spatial matrix** $S \in \mathbb{R}^{n \times n}$: which considers the real, spatial distance between atoms. In fact, S_{ij} is a monotonically decreasing function as the spatial distance between atoms i and j increases. This is because closer atoms are more likely affected by each other. Following [36], we adopt a density-like continuous function as (1):

$$S_{ij} = \begin{cases} e^{-2d^2/r^2} & 0 \leq d < r \\ \frac{4}{e^2 r^2} d^2 - \frac{12}{e^2 r} d + \frac{9}{e^2} & r \leq d < 1.5r \\ 0 & 1.5r \leq d \end{cases} \quad (1)$$

where d is the Euclidean distance between atom i and atom j , and r the average of van der Waals radii over all possible pairs of atoms. The spatial coordinates of atoms in a molecule are computed by RDKit. Note that we have $S_{ij} \in [0, 1]$.

2) *Drug Feature Learning*: Fig. 3 shows the entire procedure of Drug3D-DTI. First, the model learns a feature matrix of drug molecules from all three input (feature, bond and spatial) matrices, and then combines it with the feature matrix

TABLE I
SUMMARY OF THE BASIC ATOM FEATURES.

Feature	State
Atom type (one-hot)	C, N, O, F, P, S, Cl, Br, I, other
Degree of an atom (one-hot)	0–6
Formal charge	0 or 1
Number of radical electrons	0 or 1
Hybridization Type (one-hot)	sp, sp ² , sp ³ , sp ³ d, sp ³ d ² , other
Aromatic	0 or 1
Number of H atoms attached (one-hot)	0–4
Chirality	0 or 1
Configuration (one-hot)	R, S

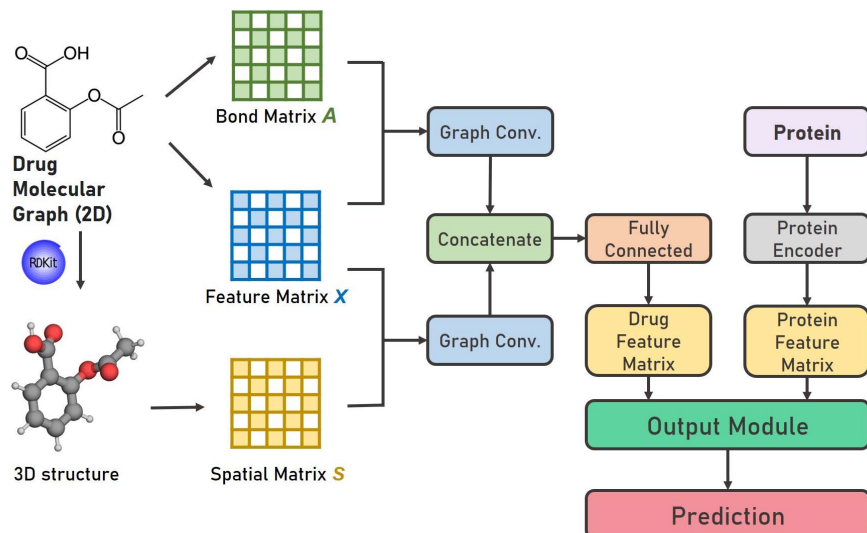


Fig. 3. The workflow of Drug3D-DTI. The 3D structure of an input molecule is first generated by using the RDKit package. From the original input and the 3D structure, three matrices $\mathbf{A}$, $\mathbf{X}$ and $\mathbf{S}$ are then generated in the preprocessing step. Two graph convolutional networks (GCNs) are trained from $\mathbf{A}$ and $\mathbf{S}$, with sharing $\mathbf{X}$. The outputs of the two GCNs are concatenated and fed into fully connected layer(s) to obtain the final feature matrix for downstream.

of target proteins to make a prediction. From the three input matrices, the first step is to use two GCNs: the pair of $\mathbf{A}$ and $\mathbf{X}$ is inputted into a GCN, while the pair of $\mathbf{S}$ and $\mathbf{X}$ is inputted into another GCN. For each atom in both GCNs, we use the following propagation rule to aggregate neighborhood information according to its chemical bond connectivity or pairwise distances in 3D space:

$$\mathbf{H}^{(l+1)} = \sigma(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)} \mathbf{W}^{(l)}) \quad (2)$$

where $\mathbf{H}^{(l)} \in \mathbb{R}^{n \times d_l}$ is the output of the l -th layer and $\mathbf{H}^{(0)}$ is $\mathbf{X}$; $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}_N$ is an updated adjacency matrix, where $\mathbf{I}_N$ is the identity matrix; $\tilde{\mathbf{D}}$ is the diagonal matrix, where $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$; $\mathbf{W}^{(l)} \in \mathbb{R}^{d_l \times d_{l+1}}$ is a trainable weight matrix of the l -th layer; σ denotes an optional activation function. For spatial matrix $\mathbf{S}$, we can just replace $\mathbf{A}$ with $\mathbf{S}$.

Note that $\tilde{\mathbf{A}}$ and corresponding $\tilde{\mathbf{D}}$ are invariant during the training process. So we can first calculate $\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}}$ in a preprocessing step and keep it all through the training process. This holds also true for $\mathbf{S}$. Further, note that the two GCNs (GCN_A for $\mathbf{A}$ and GCN_S for $\mathbf{S}$) can be parallelly computed.

In our experiments, we set the number of layers for GCN_A and GCN_S to 2 and 1, respectively.

So far, the entire computing process can be formalized as follows:

$$\mathbf{H}_A = \text{GCN}_A(\hat{\mathbf{A}}, \mathbf{X}) \quad (3)$$

$$\mathbf{H}_S = \text{GCN}_S(\hat{\mathbf{S}}, \mathbf{X}) \quad (4)$$

$$\mathbf{Z} = g(\phi(\mathbf{H}_A, \mathbf{H}_S)) \quad (5)$$

where $\mathbf{H}_A$ is from GCN_A, which captures 2D molecular features, $\mathbf{H}_S$ from GCN_S, which captures 3D structural

features, ϕ is the concatenating operation, g denotes a fully connected network, and $\mathbf{Z}$ is the final output. Each row of $\mathbf{Z}$ corresponds to the molecular feature representation of the corresponding atom. $\mathbf{Z}$ can be used for any downstream tasks as input molecular features. Note that the above description is only for a single molecule. That is, each molecule may have different numbers of atoms, and so the output matrix may have different sizes of rows. For processing multiple molecules in our experiments, we align input matrices $\mathbf{X}$, $\hat{\mathbf{A}}$ and $\hat{\mathbf{S}}$ by performing zero-padding, so that the output matrices have a common predefined size for multiple input molecules. For subsequent processes, we then use masks to avoid the influence from the "void" atoms generated by zero-padding.

B. Protein Encoder and Output Module

In this work, we apply Drug3D-DTI for drug-target interaction by using regression and binary classification.

For regression, we adopt a dual tower architecture like DeepDTA [8]. As mentioned before, the drug encoder takes into account both 2D and 3D information. The target encoder is identical to DeepDTA, a convolutional neural network operating on the target protein amino acid sequence. The output module consists of fully connected layers and dropout layers, which are also the same as DeepDTA.

For binary classification, we inherit the basic architecture of TransformerCPI [11] by replacing its drug encoder with our Drug3D-DTI drug encoder. The protein encoder of TransformerCPI is based on gated convolutional neural networks [35] with the pretrained embedding and residual connection. Well capturing internal associations between each element in a given sequence, Transformer, originally proposed for neural machine translation [34], has achieved great success in many

sequence-based learning [11], [37], [38]. The core module of Transformer is the attention mechanism, which takes three inputs: queries, keys and values. The output P can then be computed from the scaled dot product between the queries and the keys as follows:

$$P = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

where Q , K and V correspond to queries, keys and values, respectively. In this work, unlike machine translation, we removed positional encoding and decoder masking in Transformer for adapting the drug-target interaction prediction.

C. Loss Function

We use the binary cross-entropy (BCE) loss and the mean squared error (MSE) for classification and regression, respectively:

$$\text{BCE}(y, \hat{y}) = -[y \log \hat{y} + (1 - y) \log \hat{y}] \quad (7)$$

$$\text{MSE}(y, \hat{y}) = (y - \hat{y})^2 \quad (8)$$

where y is the ground truth and $\hat{y}$ is the predicted result.

IV. EXPERIMENTS

A. Datasets and Experimental Procedures

In our experiments, we used two datasets of drug-target interaction for regression and binary classification, respectively. The first one called **Davis** [39], is a kinase binding affinity dataset for regression with 30,056 pairs between 442 kinase proteins and 68 correlative inhibitors. For decreasing the influence caused by an extremely large order of magnitude, we transformed the original value of the dissociation constant (K_d) for each pair into a logarithmic scale (pK_d):

$$pK_d = -\lg K_d \quad (9)$$

Higher values of pK_d indicate stronger binding interactions. We use these logarithmic values for regression. Fig. 4 shows the distribution of these values of the Davis dataset.

The second one is the G-protein-coupled receptors (**GPCR**) dataset for binary classification [11], which consists of 15,343

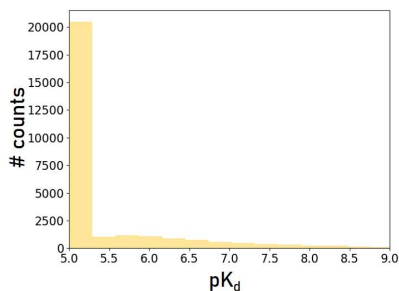


Fig. 4. Distribution of Davis kinase binding affinity dataset. [39]

TABLE II
STATISTICS OF DRUG-TARGET INTERACTION DATASETS.

Dataset	#proteins	#proteins	#pairs	#positives	#negatives
Davis	58	442	30,056	-	-
GPCR	5,359	356	15,343	7,989	7,354

GPCR-ligand binary binding affinity profiles experimentally validated between 356 proteins and 5,359 ligands.

For the Davis dataset, we conducted 5-fold cross-validation (CV) where the entire dataset was divided randomly into five-folds, with the metric being computed as the average over the five-folds. For the GPCR dataset, in order to guarantee that a model is trained from both proteins and compounds (instead of memorizing the compound patterns only), we followed the “label-reversal” training-test split described in [11]. This split strategy makes a drug appear only in one class of DTI pairs (e.g. positive) in the training set and appear only in the opposite class of DTI pairs (e.g. negative) in the test set.

Note that Davis dataset was used for the regression task, and GPCR dataset was used for the classification task. Table II summarizes the statistics of Davis and GPCR.

B. Baselines

We compared Drug3D-DTI with the four representations of drug features: 1) **ECFP**, which is exactly the 1024-bit ECFP [9]; 2) **SMILES**, which is obtained by running character-level embedding over each SMILES string; 3) **SMILES-Conv**, which is obtained by executing 1D-convolution and then max-pooling over each SMILES string [8]; and 4) **Bond-GCN**, which uses the atom feature matrix and bond matrix only [10], [11]. That is, Bond-GCN is equivalent to Drug3D-DTI without using the spatial matrix. Note that as a feature vector, ECFP was not used as the output module of Transformer.

C. Hyperparameter Settings

Table III reports hyperparameter values of Drug3D-DTI in our experiments. We used Rectified Linear Unit (ReLU) [40] as the activation function for the first GCN_A and middle fully connected layers in output modules. Only one layer was used for spatial GCN, since we consider that our 3D input matrix already captures the 3D information enough. The Adam optimizer [41] was used for updating model parameters during the training process.

D. Evaluation Metrics

For classification, we used two standard metrics: AUC (area under the receiver operating characteristic curve) and AUPR (area under the precision-recall curve). For regression, we used two common metrics: MSE (mean squared error) and PCC (Pearson correlation coefficient). For AUC, AUPR and PCC, a higher value (between 0 and 1) is better, while a lower MSE value shows better performance.

TABLE III
HYPERPARAMETER VALUES OF DRUG3D-DTI

Drug Encoder	
Hyperparameter	Value(s)
GCN hidden dimension (bond matrix A)	[128, 128]
GCN hidden dimension (spatial matrix S)	[128]
Fully connected layer units	[128]
Protein Encoder and Output Module in Dual Tower Architecture	
Hyperparameter	Value(s)
Convolutional kernels	[128, 128, 128]
Convolutional kernel size	[7, 7, 7]
Convolutional strides	[1, 1, 1]
Output fully connected layer units	[512, 128, 32, 1]
Protein Encoder and Output Module in Transformer Architecture	
Hyperparameter	Value(s)
Gated convolutional kernels	[128, 128, 128]
Gated convolutional kernel size	[7, 7, 7]
Gated convolutional strides	[1, 1, 1]
Transformer layers	3
Transformer attention heads	8
Transformer feed forward dimension	512
Output fully connected layer units	[256, 1]
Others	
Hyperparameter	Value(s)
Learning rate	1e-4
Dropout rate	0.1

E. Experimental Results and Discussion

Table IV reports the performance results of drug-target interaction regression and binary classification. In the table, the best scores in each column are highlighted in boldface. As shown, Drug3D-DTI outperforms all other compared baseline models for both tasks.

For regression on Davis with the dual tower output module, Drug3D-DTI reduces MSE by 7% and promotes PCC by 2%, compared with the original DeepDTA. SMILES obtains much lower scores than SMILES-Conv (which is used by the original DeepDTA [8]). This indicates that the convolution operation on SMILES is beneficial. Both SMILES and SMILES-Conv perform poorly than ECFP, Bond-GCN (which is used by GraphDTA [12]) and Drug3D-DTI. Note that the algorithm of ECFP can also be considered as a message-passing process on graphs. This result implies that for dual tower architecture, graph-based methods of drug molecule feature extraction are better than those based on SMILES. Among graph-based methods, Drug3D-DTI achieves the best performance, with the 4% lower MSE and 1% higher PCC compared to Bond-GCN. This confirms that the use of additional 3D structural information can enhance the model performance of drug molecule feature learning.

For binary classification, SMILES and SMILES-Conv produce the close AUC and AUPR scores around 0.840 from the output module of Transformer. This result suggests that the attention mechanism is sufficient to learn a more robust representation from 1D sequences. This is because the extra convolution operation results in no obvious improvement in classification performance. Again, graph-based methods are better than SMILES-based ones. For Transformer-

TABLE IV
RESULTS OF DRUG-TARGET INTERACTION PREDICTION.

Regression on Davis with Dual Tower Output Module		
Model	MSE	PCC
ECFP	0.216	0.856
SMILES	0.477	0.643
SMILES-Conv (DeepDTA)	0.217	0.852
Bond-GCN	0.211	0.858
Drug3D-DTI	0.202	0.867
Binary Classification on GPCR with Transformer Output Module		
Model	AUC	AUPR
ECFP	-	-
SMILES	0.849	0.840
SMILES-Conv	0.846	0.842
Bond-GCN (TransformerCPI)	0.855	0.853
Drug3D-DTI	0.867	0.865

based drug feature learning, this implies that atom-level sequences (molecular graphs) are more effective than character-level sequences (SMILES strings). In particular, Drug3D-DTI achieves the highest AUC of 0.867 and AUPR of 0.865. This outstanding performance of Drug3D-DTI demonstrates that atoms with extra 3D structural information can significantly improve prediction.

F. Hyper-parameter Analysis

Since we focus on drug feature learning for DTI prediction in this work, we examined how the dimension of the drug encoder, the numbers of GCN_A layers and the numbers of GCN_S layers affects the prediction results by varying their values while keeping other hyper-parameters unchanged. Fig. 5 depicts the model performance with different dimensions of the drug encoder, different numbers of the GCN_A layers and different numbers of the GCN_S layers. For both regression and binary classification, in the case of dimension ≤ 128 , our model benefits from the increase in the dimension. The reason for this is that longer representations are more expressive with little information loss. But too long dimensions lead to the slightly worse performance of the model because of introducing unfavorable redundancy. With 2-layer GCN_A , our model achieves the best scores on these two tasks. More or less GCN_A layers do not contribute to better prediction. This results matches the fact that organic chemical reactions often occur in the position of the first order atom and the second order atom that attach to a functional group (e.g. carbonyl alpha-substitution reactions). On the other hand, with 1-layer GCN_S our model shows the best performance, indicating that spatial interaction is direct. Therefore, we choose 128-dimension, 2-layer GCN_A and 1-layer GCN_S as the recommended hyper-parameter settings of Drug3D-DTI.

G. Case Study

We further studied the behavior of the drug encoder in Drug3D-DTI. To build a model for drug property prediction, we extracted the drug encoder and combined it to a network with multiple fully connected layers. After training this model

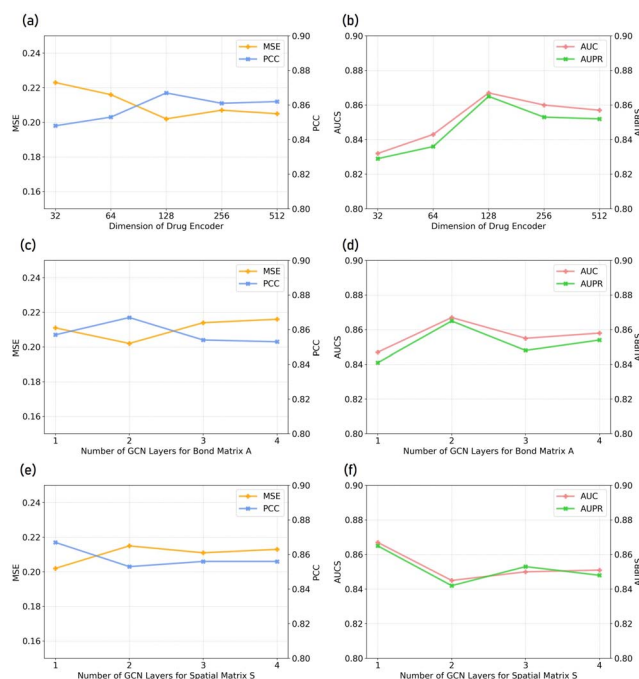


Fig. 5. Hyper-parameter analysis. (a) Regression performance with different dimensions of drug encoder on the Davis dataset. (b) Binary classification performance with different dimensions of drug encoder on the GPCR dataset. (c) Regression performance with different numbers of GCN layers (Bond) on the Davis dataset. (d) Binary classification performance with different number of GCN layers (Bond) on the GPCR dataset. (e) Regression performance with different with different number of GCN layers (Spatial) on the Davis dataset. (f) Binary classification performance with different number of GCN layers (Spatial) on the GPCR dataset.

TABLE V
CASE STUDY ON MELTING POINT PREDICTION.

SMILES-based method				
Mol.	Input SMILES string	SML	SMC	True
2-NP	<chem>O=[N+](O)c1c(O)cccc1</chem>	95.6	56.3	44.8
2-NP	<chem>C1=CC=CC(=C1O)[N+](=O)[O-]</chem>	71.8	81.3	44.8
3-NP	<chem>C1=CC=C(C=C1O)[N+](=O)[O-]</chem>	71.8	88.8	96.9
GCN-based method				
Mol.	Skeletal formula	Bond	D3D	True
2-NP	Left of Fig. 1	55.1	45.5	44.8
3-NP	Center of Fig. 1	85.6	95.4	96.9

SML: SMILES; SMC:SMILES-Conv;
Bond: Bond-GCN; D3D:Drug3D-DTI.

with a melting point dataset called Jean-Claude Bradley’s Legacy Dataset [22], we used it to predict melting points of 2-nitrophenol (2-NP) and 3-nitrophenol (3-NP), as shown in Fig. 1. Table V lists the predicted and true melting points of the two isomers. It is likely to represent the same 2-NP by two different SMILES strings. SMILES-based methods predict two different values (e.g. 95.6 and 71.8) for the 2-NP by only considering the input strings. This fact indicates the instability of such methods. In addition, one of these two SMILES strings of 2-NP is very similar to that of 3-NP. As a result, the values for 2-NP and 3-NP predicted by SMILES-based methods were very close (e.g. 71.8 for both), although the true values are

largely different (44.8 for 2-NP and 96.9 for 3-NP). This is a serious drawback of SMILES strings. In contrast, GCN-based methods are stable, with only one input for one molecule. Particularly for Bond-GCN (55.1 for 2-NP and 85.6 for 3-NP), Drug3D-DTI made more accurate predictions (45.5 for 2-NP and 95.4 for 3-NP), demonstrating the effectiveness of using the spatial information for intramolecular H-bond.

V. CONCLUSION

In this paper, we have presented the proposed Drug3D-DTI, a model that uses both 2D and 3D information of drugs with the sequence information of proteins for drug-target interaction prediction. By comparing with the competing methods, we have reported the evaluation results of the performance of Drug3D-DTI against a regression dataset, as well as a binary classification dataset. Conducting a case study, we also demonstrate that SMILES-based methods are not robust enough, while the use of additional 3D information enables Drug3D-DTI to learn drug features effectively. The 3D structures of drug molecules obtained from the chem-informatics software have been proven effective in improving predicting drug-target interaction in our experiments.

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